BCA STUDY POINT

QUESTION BANK

Semester – III Examinatation – 2026

MAJ-3

COMPUTER ARCHITECTURE

UNIT I – Introduction to Computer Architecture

2 Marks

1. What is Computer Architecture?
2. What is the importance of Computer Architecture?
3. What are the basic components of a computer system?
4. What is CPU?
5. What is ISA (Instruction Set Architecture)?
6. What is the role of ISA?
7. What is a System Bus?
8. What is an Address Bus?
9. What is a Data Bus?
10. What is a Control Bus?
11. What is a Benchmark?
12. What are Performance Metrics?
13. What is the evolution of Computer Architecture?
14. What is the difference between Hardware and Software?

5 Marks

1. ExpSlain the basic components of a computer system with a diagram.
2. Explain the role and importance of Computer Architecture in modern computing.
3. Describe the historical evolution of Computer Architecture.
4. Explain Instruction Set Architecture (ISA) and its role in defining the hardware-software interface.
5. Explain the different performance metrics and benchmarks used to evaluate computer systems.
6. Explain the architecture of a computer system with a suitable diagram.

UNIT II – Processor Architecture

2 Marks

1. What are the different types of registers used in CPU?
2. What is a Datapath?
3. What is a Control Unit?
4. What is an Instruction Cycle?
5. What is Pipelining?
6. What is Pipeline Hazard?
7. What is Instruction-Level Parallelism (ILP)?
8. What is Superscalar Architecture?
9. What is VLIW Architecture?
10. What is Branch Prediction?
11. What is Speculative Execution?
12. What is CISC Processor?
13. What is RISC Processor?
14. What is the difference between RISC and CISC?

5 Marks

1. Explain Pipelining and its principles.
2. Explain Pipeline Performance and solve numerical problems.
3. Write the fundamental differences between RISC and CISC.
4. Write any two properties of CISC Processor.
5. Explain CPU Organization and Execution Model with a diagram.
6. Explain the Instruction Execution Cycle with a suitable diagram.

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7. Explain different types of Pipeline Hazards.
8. Explain the methods used for Pipeline Performance Enhancement.
9. Explain Superscalar Architecture.
10. Explain VLIW Architecture.
11. Explain Branch Prediction and Speculative Execution.
12. Explain Instruction-Level Parallelism (ILP).

UNIT III – Memory Hierarchy and System Organization

2 Marks

1. What is Memory Hierarchy?
2. What is RAM?
3. What is ROM?
4. What is Cache Memory?
5. What is Virtual Memory?
6. Define Rotational Latency.
7. What is Seek Time?
8. What is Cache Hit?
9. What is Cache Miss?
10. What is Cache Coherence?
11. What is Locality of Reference?
12. What is Direct Mapping?
13. What is Associative Mapping?
14. What is Set-Associative Mapping?
15. What is L1 Cache?
16. Write two advantages of L1 Cache over L2 Cache.
17. What is Demand Paging?
18. What is Page Table?
19. What is MMU?
20. What is Cache Replacement Policy?
21. What is LRU?
22. What is FIFO?
23. What is Memory Consistency?
24. What is Memory Ordering?

5 Marks

1. Explain Memory Hierarchy with a diagram.
2. Explain Virtual Memory briefly.
3. A computer system has cache access time 75 ms, hit ratio 68% and average memory access time 126 ms. Find the physical memory average access time.
4. Explain the advantages and disadvantages of Cache Memory.
5. Explain Direct Mapping and solve numerical problems.
6. Explain 8-way Set Associative Mapping and find Tag Length.
7. Differentiate between Direct Mapping and Set Associative Mapping.
8. Explain Cache Coherence and Cache Replacement Policies.
9. Explain Demand Paging and Page Tables.
10. Explain Memory Management Unit (MMU).
11. Explain Cache Hit and Cache Miss.
12. Explain the principles and levels of Memory Hierarchy.
13. Explain Memory Consistency Models and Memory Ordering.

Extra Numerical Practice:

- Direct Mapped Cache: Cache size 512 KB, Block size 1 KB, Tag = 7 bits; find Main Memory Size and Tag Directory Size.
- 8-way Set Associative Cache: Cache size 32 KB, Block size 16 words, Word length 32 bits, Main Memory 4 GB; find Tag Length.
- Cache Average Access Time Numerical.
- Signed Magnitude Fraction: 16-bit pattern for -13.5 and -22.5, S = 1 bit, Exponent = 7 bits, Mantissa = 8 bits.

UNIT IV – Input/Output Systems

BEAST STUDY POINT

2 Marks

1. What is an I/O Device?
2. What is an I/O Interface?
3. What is Polling?
4. What is Interrupt?
5. What is DMA (Direct Memory Access)?
6. What is I/O Subsystem?
7. What is Bus Architecture?
8. What is Memory-Mapped I/O?
9. What is I/O-Mapped I/O?
10. What is Daisy Chaining?
11. What is I/O Performance Optimization?
12. What is Programmed I/O?

5 Marks

1. Differentiate between Memory-Mapped I/O and I/O-Mapped I/O.
2. Explain Daisy-Chaining Method with a proper diagram.
3. Explain the architecture of System Bus with a proper diagram.
4. Explain DMA (Direct Memory Access) with a diagram.
5. Explain Polling and Interrupt-driven I/O.
6. Differentiate between Polling and Interrupt.
7. Explain I/O Devices and Interfaces.
8. Explain I/O Subsystem Organization.
9. Explain different types of Bus Architecture.
10. Explain I/O Performance Optimization.

UNIT V – Parallel Architectures

2 Marks

1. What is Parallel Architecture?
2. What is Parallel Processing?
3. What is SIMD?
4. What is MIMD?
5. What is Shared Memory?
6. What is Distributed Memory?
7. What is Multiprocessor Architecture?
8. What is Multicore Architecture?
9. What is Scalability?
10. What is Synchronization?
11. What is Communication Mechanism?
12. What is Heterogeneous Computing?
13. What is Quantum Computing?
14. What is Neuromorphic Computing?

5 Marks

1. Explain the principles of Parallel Architecture.
2. Differentiate between SIMD and MIMD.
3. Explain Shared Memory and Distributed Memory Programming Models.
4. Explain Multiprocessor and Multicore Systems.
5. Explain the architecture and scalability of Multiprocessor Systems.
6. Explain Synchronization and Communication Mechanisms.
7. Write a short note on Heterogeneous Computing.
8. Explain Quantum Computing and its future applications.
9. Explain Neuromorphic Computing.
10. Explain the classification of Parallel Architectures.

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